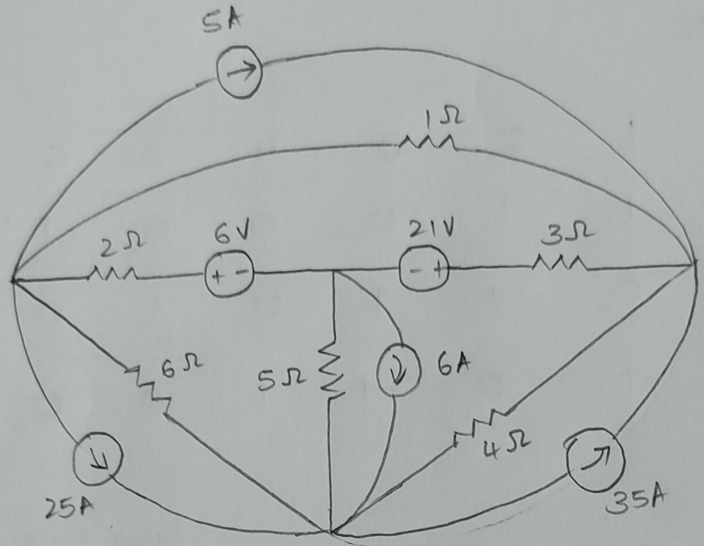
SUVARSHINI . K

MODULE 05

UNIT 02

1. For the circuit shown in figure, the ohmic values also represent the branch currents. Form a tree using 4Ω, 5Ω, 6Ω branches find the branch currents using cut set matrix



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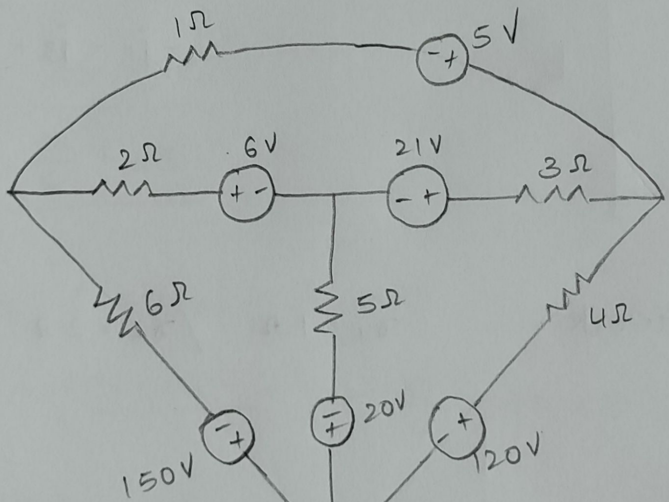
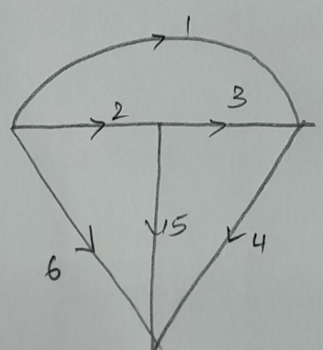
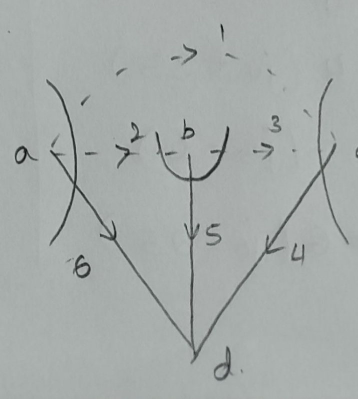
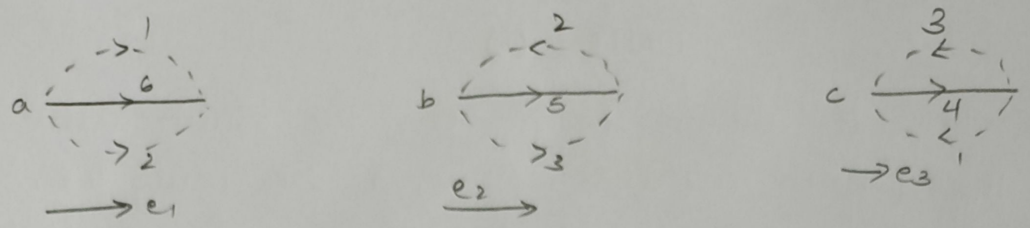


<u>OG</u>





<u>Tree</u>



cutset schedule

	1	2	3	4	5	6
e_1 (1,2,6)	1	1	0	0	0	1
e_2 (2,3,5)	0	-1	1	0	1	0
e_3 (1,3,4)	-1	0	-1	1	0	0

column wise

$$\left.\begin{aligned} v_1 &= e_1 - e_3 \\ v_2 &= e_1 - e_2 \\ v_3 &= e_2 - e_3 \\ v_4 &= e_3 \\ v_5 &= e_2 \\ v_6 &= e_1 \end{aligned}\right] \quad b$$

Row wise

$$\left.\begin{aligned} j_1 + j_2 + j_6 &= 0 \\ -j_2 + j_3 + j_5 &= 0 \\ -j_1 - j_3 + j_4 &= 0 \end{aligned}\right] \quad a$$

from the network, $r_1 = 1\Omega$ / $r_2 = 2\Omega$ / $r_3 = 3\Omega$

$$\left.\begin{aligned} v_1 &= j_1 - 5 \\ v_2 &= 2j_2 + 6 \\ v_3 &= 3j_3 - 21 \\ v_4 &= 4(j_4 + 33) \\ v_5 &= 5(j_5 - 6) \\ v_6 &= 6(j_6 - 25) \end{aligned}\right] \quad c$$

substitute c in a,

$v_1 = j_1 - 5$

$j_1 = v_1 + 5$

$\boxed{j_1 = e_1 - e_3 + 5}$

$2j_2 + 6 = e_1 - e_3$

$$\boxed{j_2 = \frac{e_1 - e_2 - 6}{2}}$$

$3j_3 - 21 = e_2 - e_3$

$$\boxed{j_3 = \frac{e_2 - e_3 + 21}{3}}$$

$4(j_4 + 35) = e_3$

$$\boxed{j_4 = \frac{e_3}{4} - 35}$$

$$\boxed{j_5 = \frac{e_2}{5} + 6}$$

$6(j_6 - 25) = e_1$

$$\boxed{j_6 = \frac{e_1}{6} + 25}$$

sub for j terms in (a)

$$e_1 - e_3 + 5 + \left(\frac{e_1 - e_2 - 6}{2}\right) + \frac{e_1}{6} + 25 = 0$$

$$e_1 - e_3 + 5 + \frac{e_1}{2} - \frac{e_2}{2} - 3 + \frac{e_1}{6} + 25 = 0$$

$$e_1(1.67) - e_2(0.5) - e_3 = -22 \rightarrow (1)$$

$$-j_2 + j_3 + j_5 = 0$$

$$-\left(\frac{e_1 - e_2 - 6}{2}\right) + \frac{e_2 - e_3 + 21}{3} + \frac{e_2}{5} + 6 = 0$$

$$-e_1(0.5) + e_2(1.033) - e_3(0.33) = -16 \rightarrow (2)$$

$$-j_1 - j_3 + j_4 = 0$$

$$-(e_1 - e_3 + 5) - \left(\frac{e_2 - e_3 + 21}{3}\right) + \left(\frac{e_3}{4} - 35\right) = 0$$

$$-e_1 - e_2(0.33) + e_3(1.583) = 47 \rightarrow (3)$$

solving 1, 2 and 3,

$e_1 = -3.537$ V

$e_2 = -8.957$ V

$e_3 = 25.57$ V

$j_1 = -24.11$ A

$j_2 = -0.26$ A

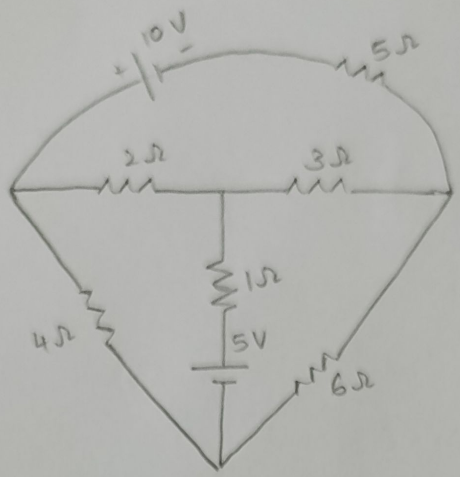
$j_3 = -4.53$ A

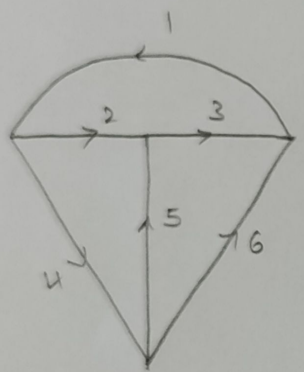
$j_4 = -28.62$ A

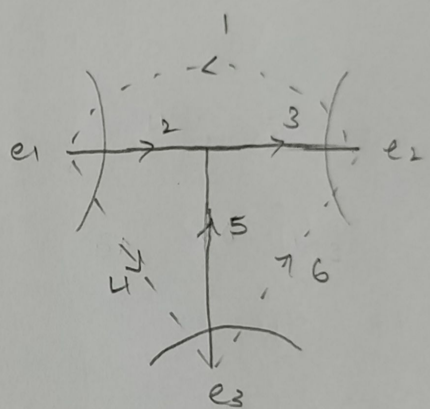
$j_5 = 4.18$ A

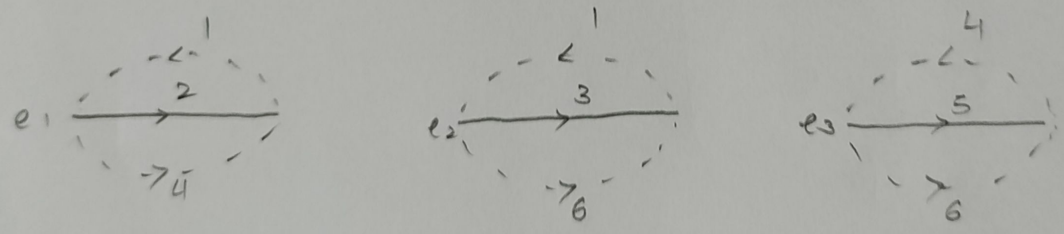
$j_6 = 24.4$ A

2. Draw the oriented graph of a network shown in figure. Select a tree, write the cutset schedule and obtain equilibrium equation.









	1	2	3	4	5	6
e_1 (1, 2, 4)	-1	01	0	1	0	0
e_2 (1, 3, 6)	1	0	-1	0	0	-1
e_3 (4, 5, 6)	0	0	0	-1	1	1

$$-j_1 + j_2 + j_4 = 0$$
$$j_1 - j_3 - j_6 = 0 \quad \text{(a) Row wise}$$
$$-j_4 + j_5 + j_6 = 0$$

Column wise,

$$V_1 = -e_1 + e_2$$
$$V_2 = e_1$$
$$V_3 = -e_2 \quad \text{(b)}$$
$$V_4 = e_1 - e_3$$
$$V_5 = e_3$$
$$V_6 = -e_2 + e_3$$

from the circuit,

$$V_1 = 5j_1 - 10$$
$$V_2 = 2j_2$$
$$V_3 = 3j_3 \quad \text{(c)}$$
$$V_4 = 4j_4$$
$$V_5 = j_5 - 5$$
$$V_6 = 6j_6$$

substitute c in b

$$j_1 = \frac{-e_1 + e_2 + 10}{5} \qquad j_4 = \frac{e_1 - e_3}{4}$$

$$j_2 = \frac{e_1}{2} \qquad j_5 = e_3 + 5$$

$$j_3 = \frac{-e_2}{3} \qquad j_6 = \frac{-e_2 + e_3}{6}$$

substitute j in terms in (a)

$$-j_1 + j_2 + j_4 = 0$$

$$\frac{e_1 - e_2 \mp 10}{5} + \frac{e_1}{2} + \frac{e_1 - e_3}{4} = 0$$

$$0.95 e_1 - 0.2 e_2 - 0.25 e_3 = -2 \rightarrow (1)$$

$$j_1 - j_3 - j_6 = 0$$

$$\frac{-e_1 + e_2 + 10}{5} + \frac{e_2}{3} + \frac{e_2 - e_3}{6} = 0$$

$$-0.2 e_1 + 0.7 e_2 - 0.166 e_3 = -2 \rightarrow (2)$$

$$-j_4 + j_5 + j_6 = 0$$

$$\frac{-e_1 + e_3}{4} + e_3 + 5 - \frac{e_2 + e_3}{6} = 0$$

$$-0.25 e_1 - 0.166 e_2 + 1.416 e_3 = -5 \rightarrow (3)$$

solving 1, 2, 3,

$e_1 = 0.287\,V$ | $V_1 = -3.987\,V$ | $j_1 = 1.202\,A$

$e_2 = -3.7\,V$ | $V_2 = 0.287\,V$ | $j_2 = 0.1435\,A$

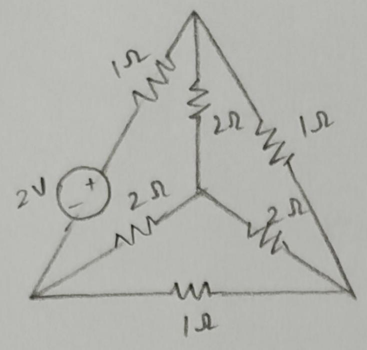
$e_3 = -3.93\,V$ | $V_3 = 3.7\,V$ | $j_3 = 1.85\,A$

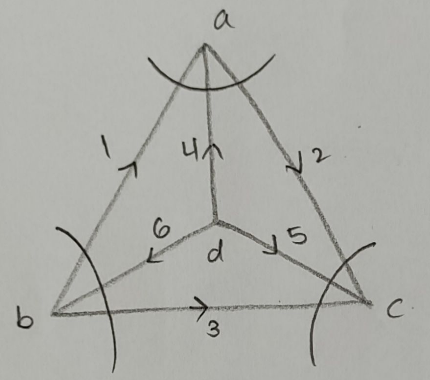
$V_4 = 4.217\,V$ | $j_4 = 1.05\,A$

$V_5 = -3.93\,V$ | $j_5 = 1.07\,A$

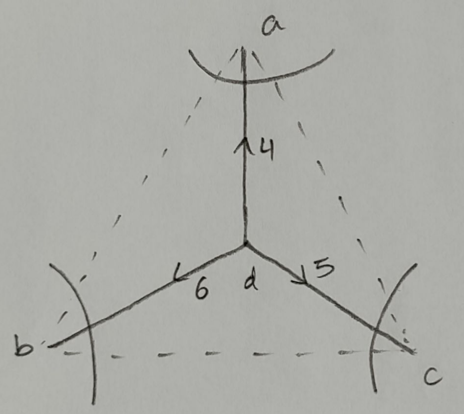
$V_6 = -0.23\,V$ | $j_6 = 0.038\,A$

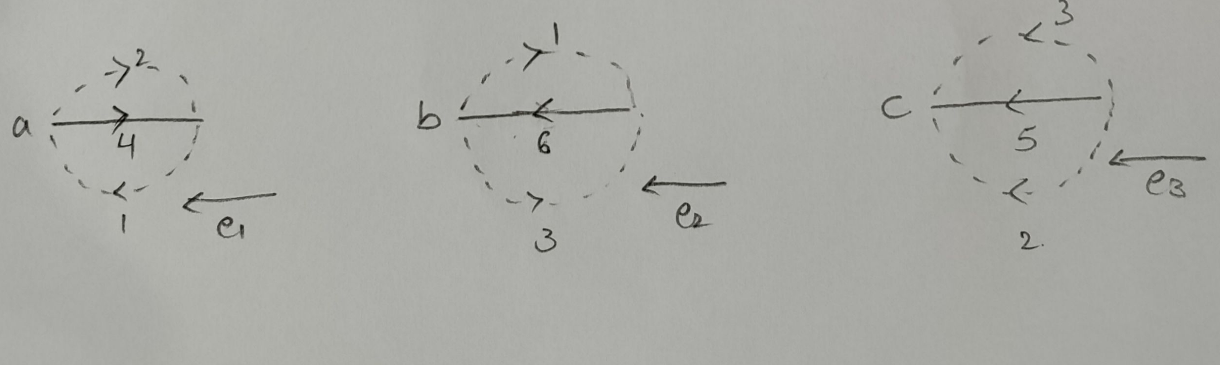
3. For this network shown in Figure, write down the cutset matrix and obtain the network equilibrium equation in matrix form using KCL.





oriented graph





				Branches		
node voltages	1	2	3	4	5	6
e_1 (1, 2, 4)	1	-1	0	1	0	0
e_2 (1, 3, 6)	-1	0	-1	0	0	1
e_3 (2, 3, 5)	0	1	1	0	1	0

Row wise $\rightarrow$ branch currents

$$j_1 - j_2 + j_4 = 0$$

$$-j_1 - j_3 + j_6 = 0$$

$$j_2 + j_3 + j_5 = 0$$

Column wise $\rightarrow$ branch voltages

$$V_1 = e_1 - e_2$$

$$V_2 = -e_1 + e_3$$

$$V_3 = -e_2 + e_3$$

$$V_4 = e_1$$

$$V_5 = e_3$$

$$V_6 = e_2$$

from the given circuit,

$r_1 = 1\,\Omega$ $\quad$ $r_4 = 2\,\Omega$

$r_2 = 1\,\Omega$ $\quad$ $r_5 = 2\,\Omega$

$r_3 = 1\,\Omega$ $\quad$ $r_6 = 2\,\Omega$

using ohm's law,

$(V = IR)$

$v_1 = j_1 r_1 = 2 \Rightarrow j_1 - 2$

$v_2 = j_2 r_2 \Rightarrow j_2$

$v_3 = j_3 r_3 \Rightarrow j_3$

$v_4 = j_4 r_4 \Rightarrow 2j_4$

$v_5 = j_5 r_5 \Rightarrow 2j_5$

$v_6 = j_6 r_6 \Rightarrow 2j_6$

Substitute eqn (3) in (2),

- $j_1 - 2 = e_1 - e_2 = e_1 - e_2 + 2.$
- $j_2 = -e_1 + e_3$
- $j_3 = -e_2 + e_3$
- $j_4 = e_1/2$
- $2j_5 = e_3/2$
- $2j_6 = e_2/2$

$j_1 - j_2 + j_4 = 0$

- $e_1 - e_2 + 2 + e_1 + e_3 + e_1/2 = 0$

$2.5e_1 - e_2 + e_3 = -2 \rightarrow (a)$

- $-e_1 - e_2 + 2 + e_2 + e_3 + \frac{e_2}{2} = 0$

$-e_1 + 2.5e_2 - e_3 = 2 \rightarrow (b)$

- $-e_1 + e_3 + (-e_2 + e_3) + \frac{e_3}{2} = 0.$

$-e_1 - e_2 + 2.5e_3 = 0 \rightarrow (c)$

Equations (a), (b) and (c) are termed as equilibrium equations.

In matrix form,

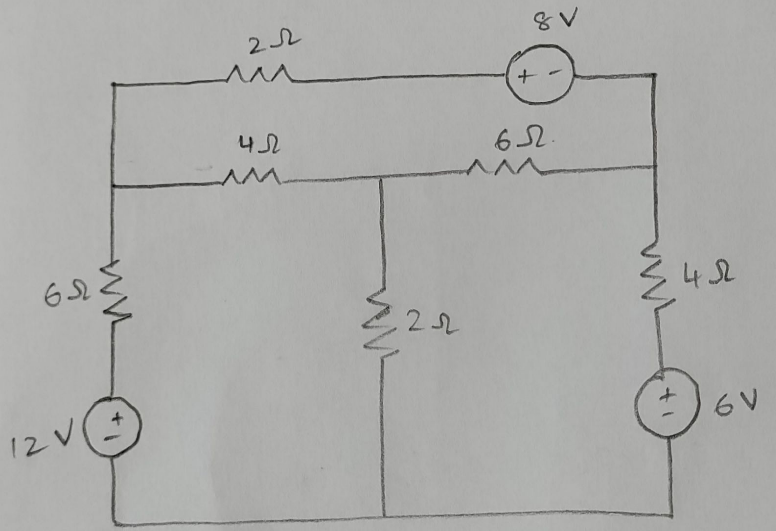
$$\begin{bmatrix} 2.5 & -1 & -1 \\ -1 & 2.5 & -1 \\ -1 & -1 & 2.5 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \\ 0 \end{bmatrix}$$

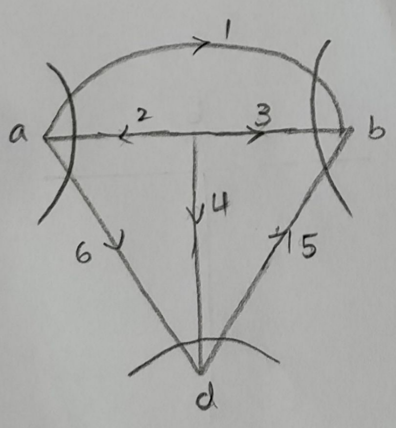
$e_1 = -0.571V$

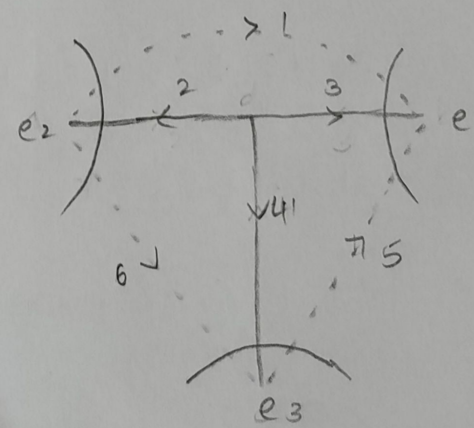
$e_2 = 0.57V$

$e_3 = 0$

4. For the network shown in Figure, write down the cutset matrix and obtain the network equilibrium equations.







	1	2	3	4	5	6
$e_1(1,3,5)$	1	0	1	0	1	0
$e_2(1,2,6)$	-1	1	0	0	0	-1
$e_3(4,5,6)$	0	0	0	1	-1	1

$$j_1 + j_3 + j_5 = 0$$

$$-j_1 + j_2 - j_6 = 0$$

$$j_4 - j_5 + j_6 = 0$$

$$V_1 = e_1 - e_2$$

$$V_2 = e_2$$

$$V_3 = e_1$$

$$V_4 = e_3$$

$$V_5 = e_1 - e_3$$

$$V_6 = -e_2 + e_3$$

$$V_1 = 2j_1 + 8$$

$$V_2 = 4j_2$$

$$V_3 = 6j_3$$

$$V_4 = 2j_4$$

$$V_5 = 4j_5 - 6$$

$$V_6 = 6j_6 + 12$$

$$j_1 = \frac{e_1 - e_2 - 8}{2}$$

$$j_2 = \frac{e_2}{4}$$

$$j_3 = \frac{e_1}{6}$$

$$j_4 = \frac{e_3}{2}$$

$$j_5 = \frac{e_1 - e_3 + 6}{4}$$

$$j_6 = \frac{-e_2 + e_3 - 12}{6}$$

substitute,

$$\frac{e_1 - e_2 - 8}{2} + \frac{e_1}{6} + \frac{e_1 - e_3 + 6}{4} = 0$$

$$0.91e_1 - 0.5e_2 - 0.25e_3 = 2.5 \rightarrow (1)$$

$$-\left(\frac{e_1 - e_2 - 8}{2}\right) + \frac{e_2}{4} - \left[\frac{-e_2 + e_3 - 12}{6}\right] = 0$$

$$-0.5e_1 + 0.91e_2 - 0.16e_3 = -6 \rightarrow (2)$$

$$0.5e_3 - \left(\frac{e_1 - e_3 + 6}{4}\right) + \left(\frac{-e_2 + e_3 - 12}{6}\right)$$

$$-0.25e_1 - 0.16e_2 + 0.91e_3 = 3.5 \rightarrow (3)$$

$e_1 = 0.277$

$e_2 = -5.93$

$e_3 = 2.87$

$V_1 = 6.207$

$V_2 = -5.93$

$V_3 = 0.277$

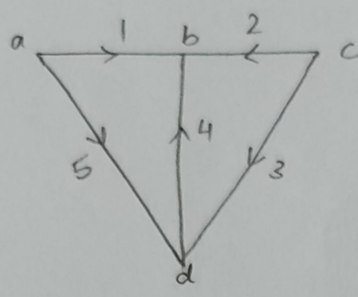
$V_4 = 2.87$

$V_5 = -2.59$

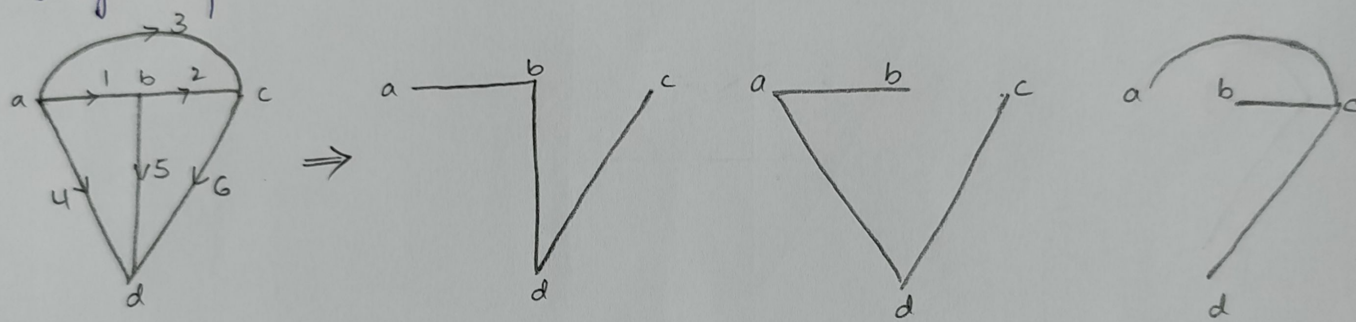
$V_6 = 8.8$

6. Explain the following with respect to Network topology

i. Oriented graph - a graph is said to be oriented when all nodes are named, ~~all its~~ and arbitary orientations are assigned to the branches which indicate the direction of branch currents.



ii. Tree - it is defined as a connected subgraph of a connected graph containing all the nodes of the graph but without any loops.



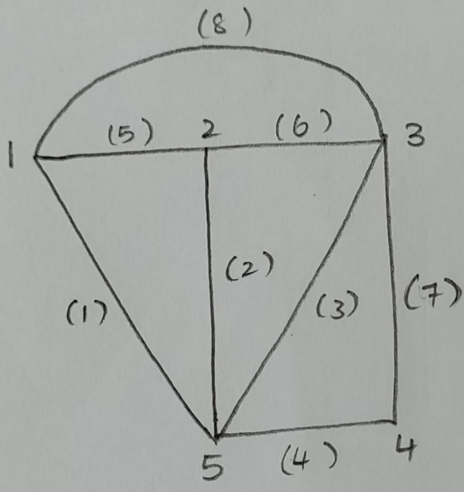
iii. Link - the branches which are not on the tree are termed as links or chords.

iv. Co-tree - The complement of tree is called co-tree which constitutes all the links.

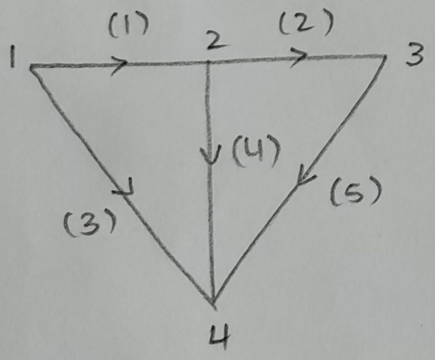
v. Connected graph - If there exists atleast one path from each node to every other node it is called connected graph.

7. Draw the oriented graph from the complete incidence matrix given below.

Nodes ↓ / Branches →	1	2	3	4	5	6	7	8
1	1	0	0	0	1	0	0	1
2	0	1	0	0	-1	1	0	0
3	0	0	1	0	0	-1	1	-1
4	0	0	0	1	0	0	-1	0
5	-1	-1	-1	-1	0	0	0	0



8. Determine and draw the number of possible trees for the graph shown.



	(1)	(2)	(3)	(4)	(5)
1	1	0	1	0	0
2	-1	1	0	1	0
3	0	-1	0	0	1
4	0	0	-1	-1	-1

$$A_{ij} = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ -1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & -1 & -1 & -1 \end{bmatrix} \leftarrow$$

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ -1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 1 \end{bmatrix}_{3\times5}$$

$$A^T = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{5\times3}$$

$$A \cdot A^T = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 3 & -1 \\ 0 & -1 & 2 \end{bmatrix} = 8$$